

1. What type of operating system is suitable for safety-critical systems and why?
2. Should browser tabs be implemented as threads or separate processes? Justify your answer.
3. Which CPU scheduling approach is suitable for interactive multi-user environments?
4. Which IPC mechanism is appropriate for transferring large volumes of data quickly?
5. Why do race conditions occur and how can they be prevented?
6. How is the producer–consumer problem solved using synchronization?
7. How does the Readers–Writers approach optimize performance?
8. What is deadlock and how can it be prevented?
9. What causes thrashing and how can it be corrected?
10. How can programs run when they require more memory than available RAM?
11. What is hardware abstraction in operating systems?
12. Why do operating systems enforce process isolation?
13. What is mutual exclusion and why is it required?
14. What conditions must a correct critical section solution satisfy?
15. Compare binary and counting semaphores.
16. What is the purpose of a Resource Allocation Graph?
17. Compare contiguous and non-contiguous memory allocation.
18. Differentiate internal and external fragmentation.
19. How does lock granularity affect performance?
20. Why is OS design considered a trade-off problem?
21. How does CI prevent issues caused by manual deployment?
22. What is Continuous Delivery in CI/CD pipelines?
23. Why are webhooks preferred over polling in Jenkins?
24. Why does running mvn package trigger compilation and testing?
25. How does Kubernetes maintain replica counts after failure?
26. What is Continuous Integration and its goal?
27. What is a Jenkins pipeline?
28. What is the purpose of pom.xml in Maven?
29. What is a container image?
30. Why is artifact versioning important?
31. What role do Jenkins plugins play?
32. How does Maven resolve transitive dependencies?
33. What is the role of a Kubernetes service?
34. Why is declarative infrastructure management important?
35. How does monitoring improve deployment reliability?
36. Why can accuracy be misleading in imbalanced datasets?
37. Explain the precision–recall trade-off.
38. What causes overfitting and how can it be addressed?
39. What causes underfitting and how can it be corrected?
40. What checks should be done before deploying an ML model?
41. What are the components of a confusion matrix?
42. Define True Positive and False Negative.
43. What is precision in model evaluation?
44. What is recall and when is it important?
45. Why should datasets be split into training, validation, and test sets?
46. What is cross-validation used for?
47. Why is threshold selection important?

48. Why must evaluation metrics match the problem type?
49. Why should multiple metrics be considered during evaluation?
50. What is starvation in Round Robin scheduling?
51. Give a real-world computing example of the Dining Philosophers problem.
52. Is address translation only between logical and physical addresses? Explain.
53. What is one approach to reduce internal and external fragmentation?
54. How do system calls differ across operating systems?
55. What is the difference between a container image and a Docker image?